```
# Mountung google drive
from google.colab import drive

drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
# Importing the necessary libraries
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import seaborn as sns
import os
import numpy as np
```

```
# Loading the Essential datasets
orders_df = pd.read_csv('/content/drive/MyDrive/Amali/olist_orders_dataset.csv')
reviews_df = pd.read_csv('/content/drive/MyDrive/Amali/olist_order_reviews_dataset.csv')
customers_df = pd.read_csv('/content/drive/MyDrive/Amali/olist_customers_dataset.csv')
products_df = pd.read_csv('/content/drive/MyDrive/Amali/olist_products_dataset.csv')
```

```
# Loading helper dataset
order_items_df = pd.read_csv('/content/drive/MyDrive/Amali/olist_order_items_dataset.csv')
print(order_items_df.shape)
print(order_items_df.columns.tolist())
```

```
(112650, 7)
['order_id', 'order_item_id', 'product_id', 'seller_id', 'shipping_limit_date', 'price', 'freight_value']
```

Inspecting the dataframes

```
orders_df.head()
```

	order_id	customer_id	order_status	order_purchase_timestamp	order_approved_at	order_delivered_carrier_date	order_delivered_customer_dat
0	e481f51cbdc54678b7cc49136f2d6af7	9ef432eb6251297304e76186b10a928d	delivered	2017-10-02 10:56:33	2017-10-02 11:07:15	2017-10-04 19:55:00	2017-10-10 21:25:1
1	53cdb2fc8bc7dce0b6741e2150273451	b0830fb4747a6c6d20dea0b8c802d7ef	delivered	2018-07-24 20:41:37	2018-07-26 03:24:27	2018-07-26 14:31:00	2018-08-07 15:27:4
2	47770eb9100c2d0c44946d9cf07ec65d	41ce2a54c0b03bf3443c3d931a367089	delivered	2018-08-08 08:38:49	2018-08-08 08:55:23	2018-08-08 13:50:00	2018-08-17 18:06:2
3	949d5b44dbf5de918fe9c16f97b45f8a	f88197465ea7920adcdbec7375364d82	delivered	2017-11-18 19:28:06	2017-11-18 19:45:59	2017-11-22 13:39:59	2017-12-02 00:28:4
4	ad21c59c0840e6cb83a9ceb5573f8159	8ab97904e6daea8866dbdbc4fb7aad2c	delivered	2018-02-13 21:18:39	2018-02-13 22:20:29	2018-02-14 19:46:34	2018-02-16 18:17:0

Next steps: Generate code with orders_df New interactive sheet

```
reviews_df.head()
```

	review_id	order_id	review_score	review_comment_title	review_comment_message	review_creation_date	review_answer_timestamp
0	7bc2406110b926393aa56f80a40eba40	73fc7af87114b39712e6da79b0a377eb	4	NaN	NaN	2018-01-18 00:00:00	2018-01-18 21:46:59
1	80e641a11e56f04c1ad469d5645fdfde	a548910a1c6147796b98fdf73dbeba33	5	NaN	NaN	2018-03-10 00:00:00	2018-03-11 03:05:13
2	228ce5500dc1d8e020d8d1322874b6f0	f9e4b658b201a9f2ecdecbb34bed034b	5	NaN	NaN	2018-02-17 00:00:00	2018-02-18 14:36:24
3	e64fb393e7b32834bb789ff8bb30750e	658677c97b385a9be170737859d3511b	5	NaN	Recebi bem antes do prazo estipulado.	2017-04-21 00:00:00	2017-04-21 22:02:06

Next steps: Generate code with reviews_df New interactive sheet

```
customers_df.head()
```

	customer_id	customer_unique_id	customer_zip_code_prefix	customer_city	customer_state
0	06b8999e2fba1a1fbc88172c00ba8bc7	861eff4711a542e4b93843c6dd7febb0	14409	franca	SP
1	18955e83d337fd6b2def6b18a428ac77	290c77bc529b7ac935b93aa66c333dc3	9790	sao bernardo do campo	SP
2	4e7b3e00288586ebd08712fdd0374a03	060e732b5b29e8181a18229c7b0b2b5e	1151	sao paulo	SP
3	b2b6027bc5c5109e529d4dc6358b12c3	259dac757896d24d7702b9acbbff3f3c	8775	mogi das cruzes	SP
4	4f2d8ab171c80ec8364f7c12e35b23ad	345ecd01c38d18a9036ed96c73b8d066	13056	campinas	SP

Next steps: Generate code with customers_df New interactive sheet

```
products_df.head()
```

	product_id	product_category_name	product_name_lenght	product_description_lenght	product_photos_qty	product_weight_g	product_length_cm	product_height_cm	produc
0	1e9e8ef04dbcff4541ed26657ea517e5	perfumaria	40.0	287.0	1.0	225.0	16.0	10.0	
1	3aa071139cb16b67ca9e5dea641aaa2f	artes	44.0	276.0	1.0	1000.0	30.0	18.0	
2	96bd76ec8810374ed1b65e291975717f	esporte_lazer	46.0	250.0	1.0	154.0	18.0	9.0	
3	cef67bcfe19066a932b7673e239eb23d	bebes	27.0	261.0	1.0	371.0	26.0	4.0	
4	9dc1a7de274444849c219cff195d0b71	utilidades_domesticas	37.0	402.0	4.0	625.0	20.0	17.0	

Next steps: Generate code with products_df New interactive sheet

```
# The orders dataset
print(orders_df.shape)
print(orders_df.isnull().sum())
print(orders_df['order_status'].value_counts())
```

```
(99441, 8)
order_id                      0
customer_id                   0
order_status                  0
order_purchase_timestamp      0
order_approved_at           160
order_delivered_carrier_date 1783
```

```
order_delivered_customer_date    2965
order_estimated_delivery_date       0
dtype: int64
order_status
delivered      96478
shipped         1107
canceled         625
unavailable      609
invoiced         314
processing       301
created            5
approved           2
Name: count, dtype: int64
```

```
# The reviews dataset
print(reviews_df.shape)
print(reviews_df.isnull().sum())
print(reviews_df['review_score'].value_counts().sort_index())
```

```
(99224, 7)
review_id                     0
order_id                      0
review_score                  0
review_comment_title      87656
review_comment_message    58247
review_creation_date          0
review_answer_timestamp       0
dtype: int64
review_score
1    11424
2     3151
3     8179
4    19142
5    57328
Name: count, dtype: int64
```

```
# The customers dataset
print(customers_df.shape)
print(customers_df.isnull().sum())
```

```
(99441, 5)
customer_id               0
customer_unique_id        0
customer_zip_code_prefix  0
customer_city             0
customer_state            0
dtype: int64
```

```
# The products dataset
print(products_df.shape)
print(products_df.isnull().sum())
```

```
(32951, 9)
product_id                    0
product_category_name       610
product_name_lenght         610
product_description_lenght  610
product_photos_qty          610
product_weight_g              2
product_length_cm             2
product_height_cm             2
product_width_cm              2
dtype: int64
```

```
print('\n Unique values')
print (orders_df["order_id"].nunique() == len(orders_df))
print (reviews_df["order_id"].nunique() == len(reviews_df))
print (customers_df["customer_id"].nunique() == len(customers_df))
```

```
 Unique values
True
False
True
```

Joining The Tables

```
# Joining Orders + Customers on customer_id
master_df = orders_df.merge(customers_df, on='customer_id', how='left')
print(f"After orders + customers: {master_df.shape}")

# STEP 3: Join + Review on order_id
master_df = master_df.merge(reviews_df, on='order_id', how='left')
print(f"After + reviews: {master_df.shape}")

# STEP 4: Joining + Order Items ( the bridge dataset) and then + Products
# Agreggating order_items to one row per order BEFORE joining
order_items_agg = order_items_df.groupby('order_id').agg(
    product_id=('product_id', 'first'),
    order_item_count=('order_item_id', 'count')
).reset_index()

master_df = master_df.merge(order_items_agg, on='order_id', how='left')
print(f"After + order_items: {master_df.shape}")

# Joining then to Products on product_id
master_df = master_df.merge(products_df[['product_id', 'product_category_name']],
                            on='product_id', how='left')
print(f"After + products: {master_df.shape}")

print("\nFinal columns:")
print(master_df.columns.tolist())
```

```
After orders + customers: (99441, 12)
After + reviews: (99992, 18)
After + order_items: (99992, 20)
After + products: (99992, 21)

Final columns:
['order_id', 'customer_id', 'order_status', 'order_purchase_timestamp', 'order_approved_at', 'order_delivered_carrier_date', 'order_delivered_customer_date', 'order_estimated_delivery_date', 'customer_unique_i
```

Cleaning The Resulting Dataset For Reliability

```
# Fixing duplicate rows due to one to many reviews (keeping latest review per order)

# First, converting review_creation_date to datetime so we can sort it
```

```
master_df['review_creation_date'] = pd.to_datetime(master_df['review_creation_date'])

# Sort by order_id and review date, keep the most recent review
master_df = master_df.sort_values('review_creation_date', ascending=False)
master_df = master_df.drop_duplicates(subset='order_id', keep='first')

print(f"After removing duplicates: {master_df.shape}")
```

```
After removing duplicates: (99441, 21)
```

```
# Fixing orders where the status is not delivered since they can't be classified in relation to our problem

# First, filtering to delivered orders only
master_df = master_df[master_df['order_status'] == 'delivered'].copy()
print(f"Delivered orders only: {master_df.shape}")

# Then, converting date columns to datetime
date_cols = [
    'order_purchase_timestamp',
    'order_approved_at',
    'order_delivered_carrier_date',
    'order_delivered_customer_date',
    'order_estimated_delivery_date'
]

for col in date_cols:
    master_df[col] = pd.to_datetime(master_df[col])

print("Date columns converted:")
print(master_df[date_cols].dtypes)

# Dropping rows where delivery date is null
master_df = master_df.dropna(subset=['order_delivered_customer_date'])
print(f"After dropping null delivery dates: {master_df.shape}")

# Calculating delay in days
master_df['days_difference'] = (
    master_df['order_estimated_delivery_date'] -
    master_df['order_delivered_customer_date']
).dt.days

print("\nDelay stats:")
print(master_df['days_difference'].describe())

# Classifying delivery status
def classify_delivery(days):
    if days >= 0:
        return 'On Time'
    elif days >= -5:
        return 'Late'
    else:
        return 'Super Late'

master_df['delivery_status'] = master_df['days_difference'].apply(classify_delivery)

print("\nDelivery status breakdown:")
print(master_df['delivery_status'].value_counts())
```

```
Delivered orders only: (96478, 21)
Date columns converted:
order_purchase_timestamp         datetime64[ns]
order_approved_at                datetime64[ns]
order_delivered_carrier_date     datetime64[ns]
order_delivered_customer_date    datetime64[ns]
order_estimated_delivery_date    datetime64[ns]
dtype: object
After dropping null delivery dates: (96470, 21)

Delay stats:
count    96470.000000
mean        10.875889
std         10.182105
min       -189.000000
25%          6.000000
50%         11.000000
75%         16.000000
max        146.000000
Name: days_difference, dtype: float64

Delivery status breakdown:
delivery_status
On Time       88644
Super Late     4211
Late           3615
Name: count, dtype: int64
```

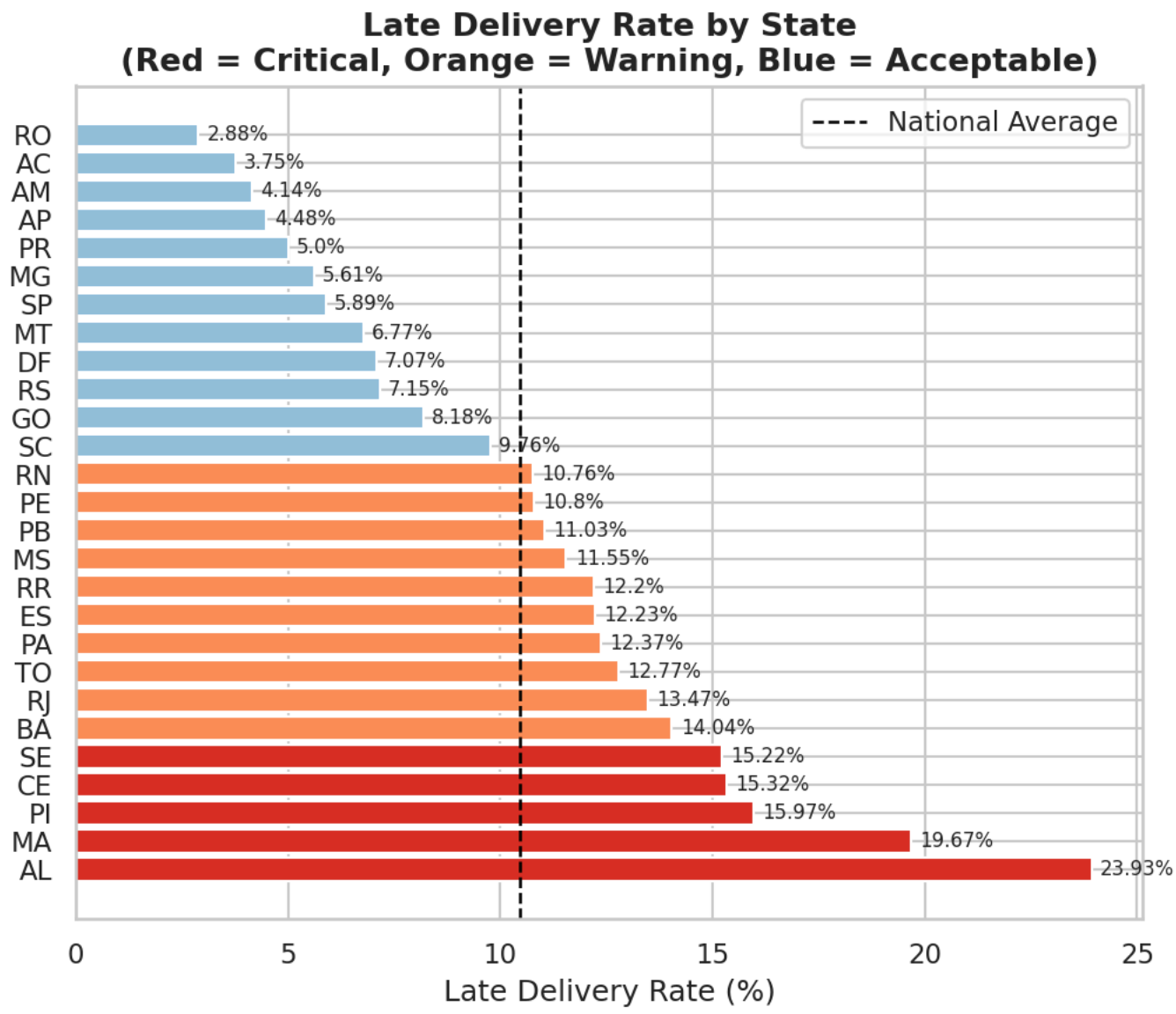
```
# GEOGRAPHIC ANALYSIS: Late % per State

# Creating a flag for late orders
master_df['is_late'] = master_df['delivery_status'].isin(['Late', 'Super Late']).astype(int)

# Aggregating by state
state_analysis = master_df.groupby('customer_state').agg(
    total_orders=('order_id', 'count'),
    late_orders=('is_late', 'sum'),
    avg_review_score=('review_score', 'mean'),
    avg_delay_days=('days_difference', 'mean')
).reset_index()

# Calculating late percentage
state_analysis['late_pct'] = (
    state_analysis['late_orders'] / state_analysis['total_orders'] * 100
).round(2)

# Sorting by worst performing states
state_analysis = state_analysis.sort_values('late_pct', ascending=False)

print(state_analysis.to_string())
```

```
   customer_state  total_orders  late_orders  avg_review_score  avg_delay_days  late_pct
1              AL           397           95          3.847716        7.707809     23.93
9              MA           717          141          3.832865        8.571827     19.67
16             PI           476           76          3.993631       10.306723     15.97
5              CE          1279          196          3.944226        9.804535     15.32
24             SE           335           51          3.907186        9.020896     15.22
4              BA          3256          457          3.929700        9.794533     14.04
18             RJ         12350         1664          3.965687       10.761215     13.47
26             TO           274           35          4.153846       11.131387     12.77
13             PA           946          117          3.914255       13.066596     12.37
7              ES          1995          244          4.078720        9.496241     12.23
21             RR            41            5          3.902439       16.292683     12.20
11             MS           701           81          4.161660       10.052782     11.55
```

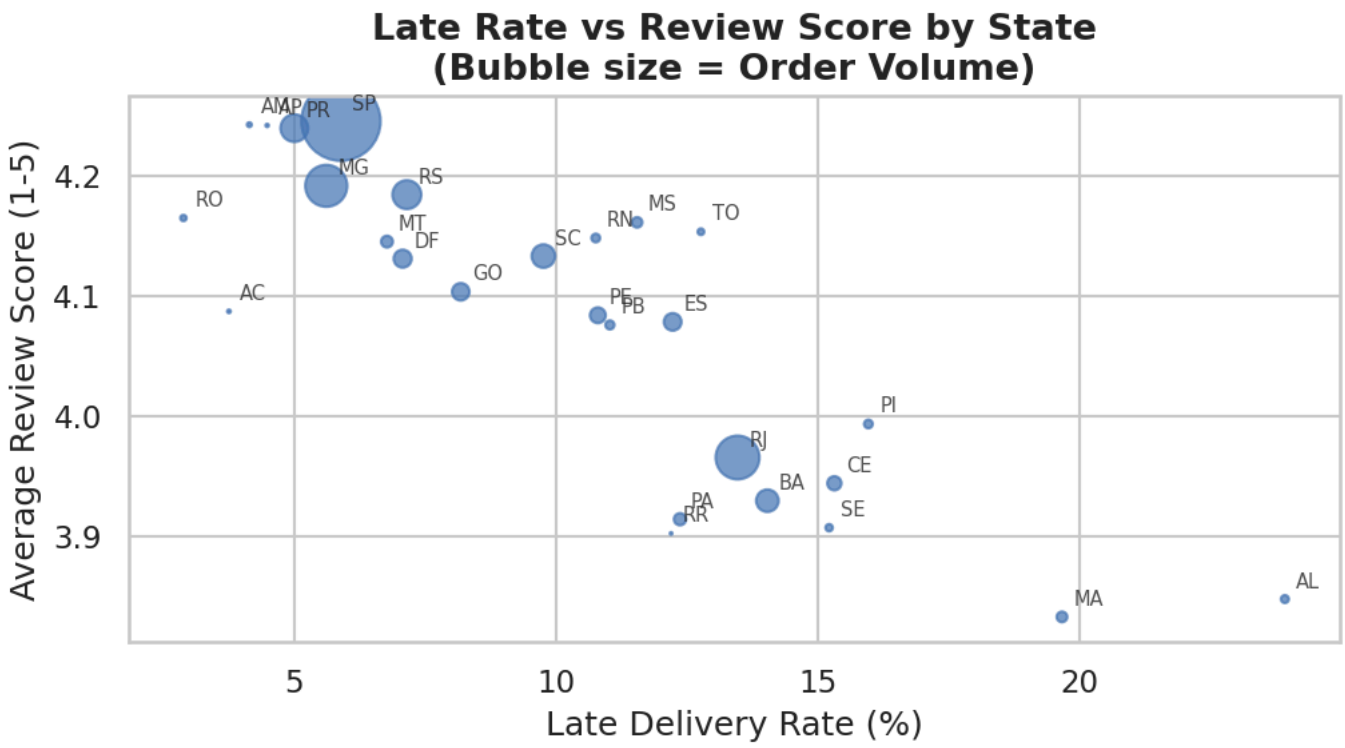
```
14          PB        517        57       4.076172      12.261122     11.03
15          PE       1593       172       4.084231      12.293785     10.80
19          RN        474        51       4.148620      12.649789     10.76
23          SC       3546       346       4.133561      10.503384      9.76
8           GO       1957       160       4.103803      11.185488      8.18
22          RS       5344       382       4.184754      12.910367      7.15
6           DF       2080       147       4.131401      11.048077      7.07
12          MT        886        60       4.145620      13.363431      6.77
25          SP      40494      2387       4.245940      10.075542      5.89
10          MG      11354       637       4.192113      12.242998      5.61
17          PR       4923       246       4.240204      12.314239      5.00
3           AP         67         3       4.242424      18.686567      4.48
2           AM        145         6       4.243056      18.565517      4.14
0           AC         80         3       4.087500      19.725000      3.75
20          RO        243         7       4.165289      19.102881      2.88
```

Plotting The graphs

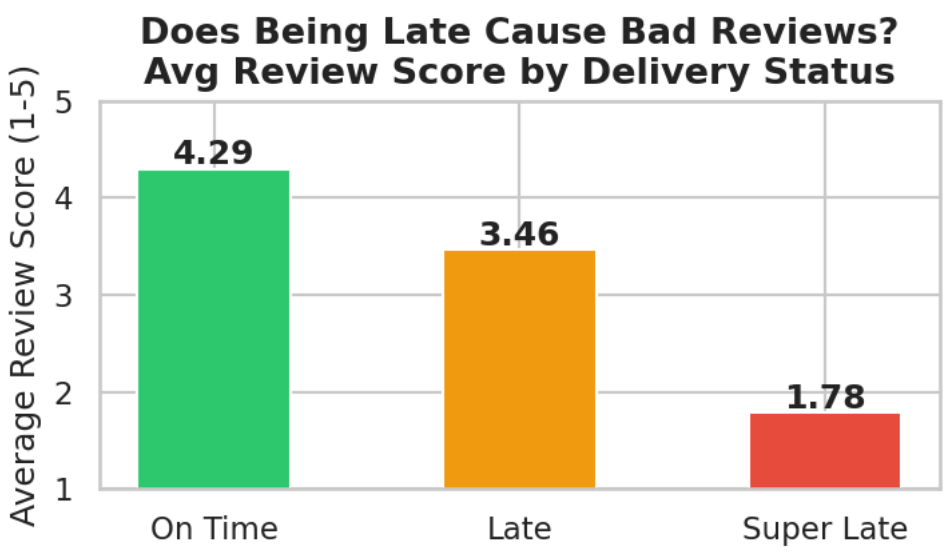
```
# CHART 1: Late % by State (Horizontal Bar)

fig, ax = plt.subplots(figsize=(7, 6))

colors = ['#d73027' if x > 15 else '#fc8d59' if x > 10 else '#91bfdb'
          for x in state_analysis['late_pct']]

bars = ax.barh(state_analysis['customer_state'],
               state_analysis['late_pct'],
               color=colors)

ax.axvline(x=state_analysis['late_pct'].mean(),
           color='black', linestyle='--', linewidth=1.2, label='National Average')

ax.set_xlabel('Late Delivery Rate (%)')
ax.set_title('Late Delivery Rate by State\n(Red = Critical, Orange = Warning, Blue = Acceptable)',
             fontsize=13, fontweight='bold')
ax.legend()

for bar, val in zip(bars, state_analysis['late_pct']):
    ax.text(val + 0.2, bar.get_y() + bar.get_height()/2,
            f'{val}%', va='center', fontsize=8)

plt.tight_layout()
plt.savefig('chart1_late_by_state.png', bbox_inches='tight')
plt.show()
```



```
# CHART 2: Late % vs Average Review Score

fig, ax = plt.subplots(figsize=(7, 4))

scatter = ax.scatter(state_analysis['late_pct'],
                     state_analysis['avg_review_score'],
                     s=state_analysis['total_orders']/50,
                     alpha=0.7,
                     color='#4575b4')

# Label each bubble with state name
for _, row in state_analysis.iterrows():
    ax.annotate(row['customer_state'],
                (row['late_pct'], row['avg_review_score']),
                fontsize=7, alpha=0.8,
                xytext=(4, 4), textcoords='offset points')

ax.set_xlabel('Late Delivery Rate (%)')
ax.set_ylabel('Average Review Score (1-5)')
ax.set_title('Late Rate vs Review Score by State\n(Bubble size = Order Volume)',
             fontsize=13, fontweight='bold')

plt.tight_layout()
plt.savefig('chart2_late_vs_review.png', bbox_inches='tight')
plt.show()
```

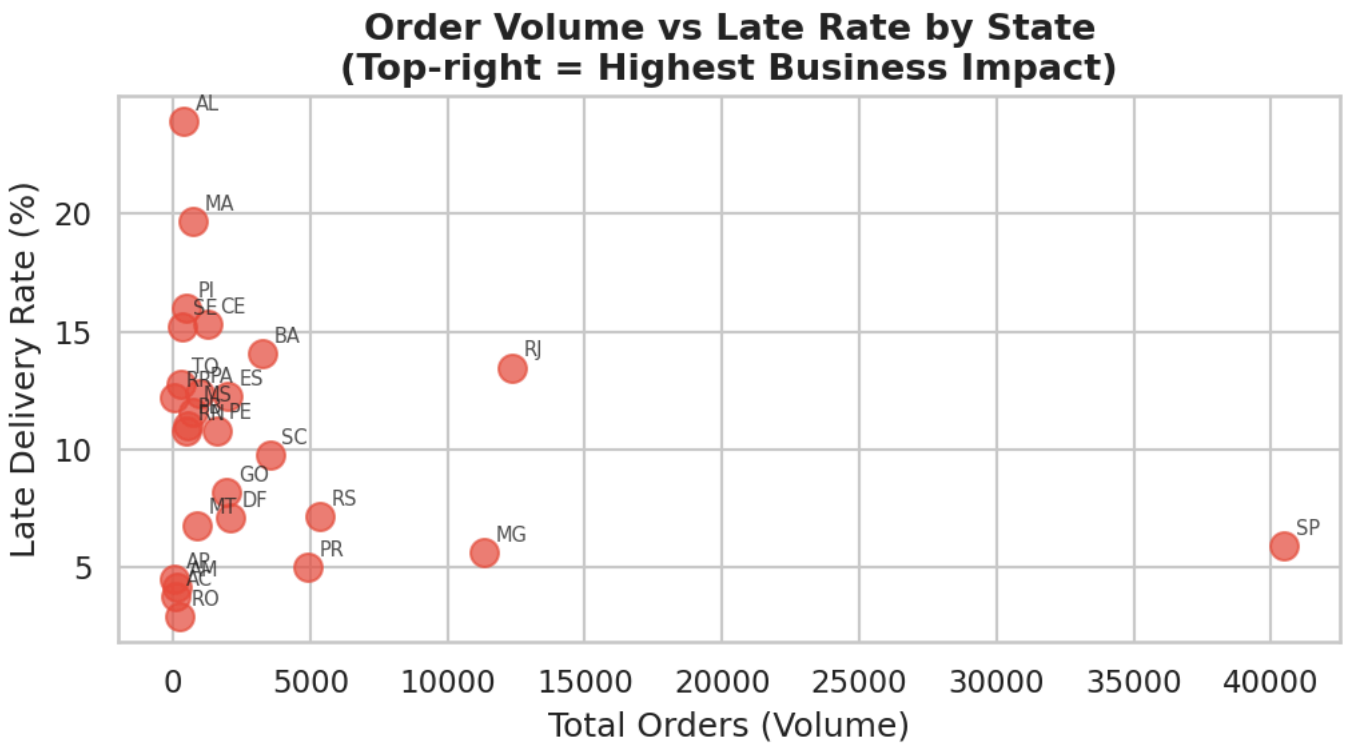



```
# CHART 3: Average Review Score by Delivery Status

status_review = master_df.groupby('delivery_status')['review_score'].mean().reset_index()
status_order = ['On Time', 'Late', 'Super Late']
status_review['delivery_status'] = pd.Categorical(
    status_review['delivery_status'], categories=status_order, ordered=True)
status_review = status_review.sort_values('delivery_status')

fig, ax = plt.subplots(figsize=(5, 3))

colors = ['#2ecc71', '#f39c12', '#e74c3c']
bars = ax.bar(status_review['delivery_status'],
              status_review['review_score'],
              color=colors, width=0.5)

ax.set_ylim(1, 5)
ax.set_ylabel('Average Review Score (1-5)')
ax.set_title('Does Being Late Cause Bad Reviews?\nAvg Review Score by Delivery Status',
             fontsize=13, fontweight='bold')

for bar, val in zip(bars, status_review['review_score']):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.05,
            f'{val:.2f}', ha='center', fontweight='bold')

plt.tight_layout()
plt.savefig('chart3_review_by_status.png', bbox_inches='tight')
plt.show()
```




```
# CHART 4: Order Volume vs Late Rate

fig, ax = plt.subplots(figsize=(7, 4))

ax.scatter(state_analysis['total_orders'],
           state_analysis['late_pct'],
           s=100, alpha=0.7, color='#e74c3c')

for _, row in state_analysis.iterrows():
```

```
    ax.annotate(row['customer_state'],
                (row['total_orders'], row['late_pct']),
                fontsize=7, alpha=0.8,
                xytext=(4, 4), textcoords='offset points')

ax.set_xlabel('Total Orders (Volume)')
ax.set_ylabel('Late Delivery Rate (%)')
ax.set_title('Order Volume vs Late Rate by State\n(Top-right = Highest Business Impact)',
             fontsize=13, fontweight='bold')

plt.tight_layout()
plt.savefig('chart4_volume_vs_laterate.png', bbox_inches='tight')
plt.show()
```



Candidate's Choice: ANOMALY DETECTION: Monthly Late Rate by State

```
# Extracting year-month from purchase timestamp
master_df['order_month'] = master_df['order_purchase_timestamp'].dt.to_period('M')

# Calculating late rate per state per month
monthly_state = master_df.groupby(['customer_state', 'order_month']).agg(
    total_orders=('order_id', 'count'),
    late_orders=('is_late', 'sum')
).reset_index()

monthly_state['late_pct'] = (
    monthly_state['late_orders'] / monthly_state['total_orders'] * 100
).round(2)

# Checking how many months of data each state has
months_per_state = monthly_state.groupby('customer_state')['order_month'].count()
print("Months of data per state:")
print(months_per_state.sort_values(ascending=False).to_string())

# Checking average orders per month per state
avg_orders_per_month = monthly_state.groupby('customer_state')['total_orders'].mean()
print("\nAvg orders per month per state:")
print(avg_orders_per_month.sort_values(ascending=False).round(1).to_string())
```

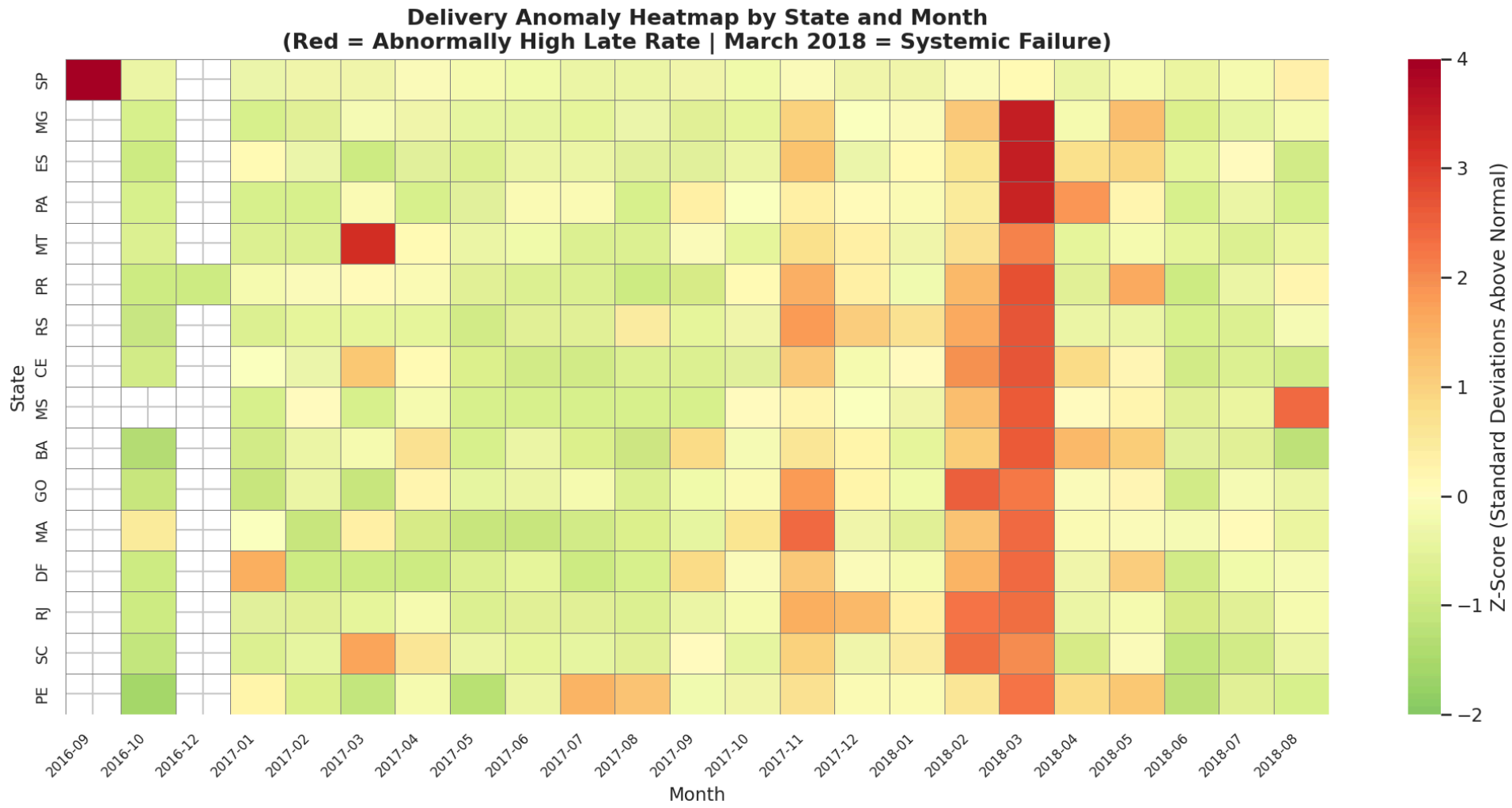
```
Months of data per state:
customer_state
PR    22
SP    22
AL    21
MA    21
GO    21
DF    21
ES    21
MG    21
PB    21
CE    21
BA    21
SE    21
RN    21
SC    21
MT    21
PI    21
PE    21
RJ    21
PA    21
RS    21
MS    20
AC    20
RO    20
TO    20
AM    19
AP    18
RR    17

Avg orders per month per state:
customer_state
SP    1840.6
RJ     588.1
MG     540.7
RS     254.5
PR     223.8
SC     168.9
BA     155.0
DF      99.0
ES      95.0
GO      93.2
PE      75.9
CE      60.9
PA      45.0
MT      42.2
MS      35.0
MA      34.1
PB      24.6
PI      22.7
```

```
RN    22.6
AL    18.9
SE    16.0
TO    13.7
RO    12.2
AM     7.6
AC     4.0
```

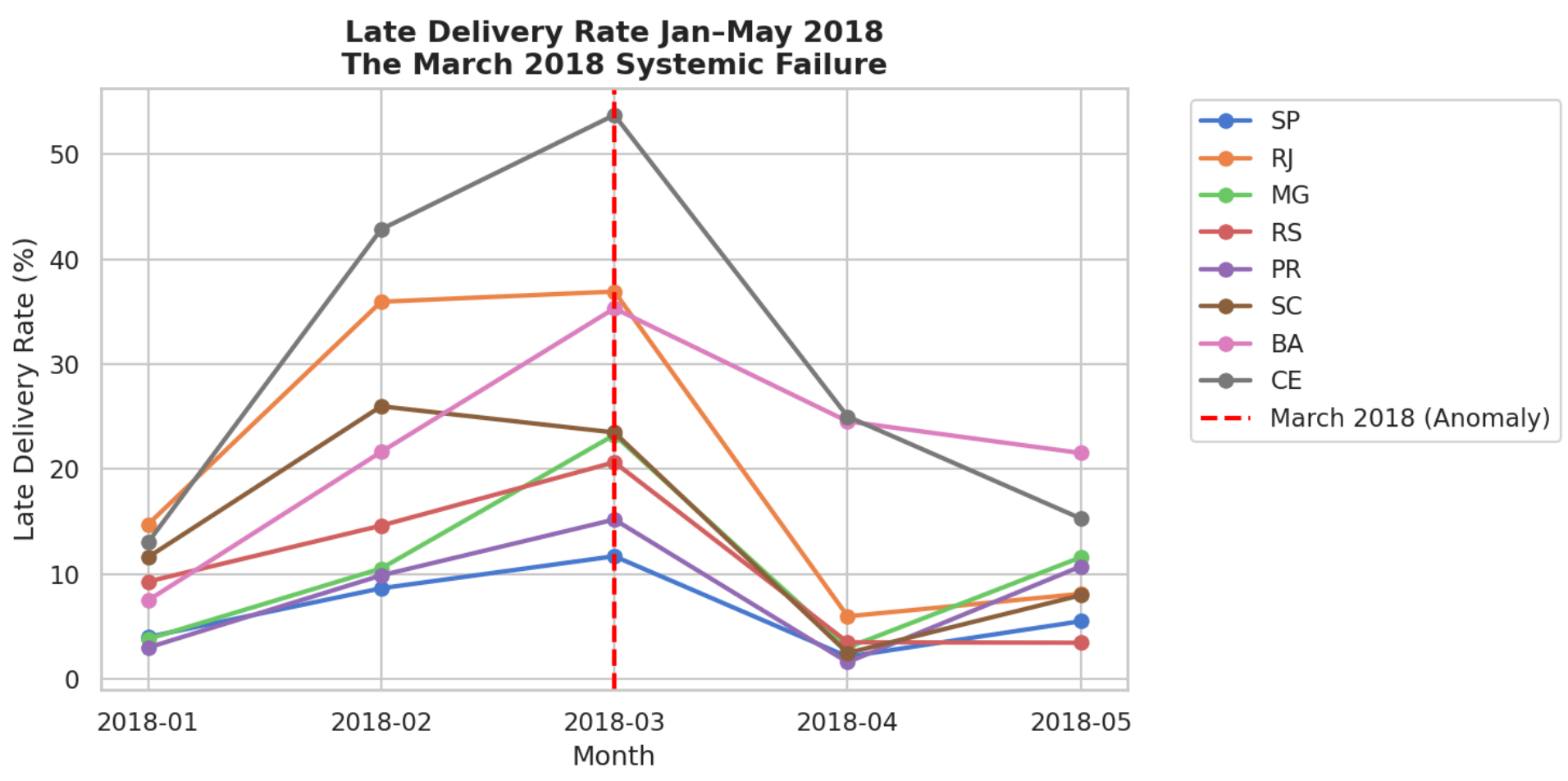
```
# Maintaining Integrity of the data acquired in the first step

# Filter to states with enough data
sufficient_states = avg_orders_per_month[avg_orders_per_month >= 30].index.tolist()
monthly_filtered = monthly_state[monthly_state['customer_state'].isin(sufficient_states)].copy()

print(f"States included: {sufficient_states}")

# Calculate mean and std per state ( each state's normal range)
state_stats = monthly_filtered.groupby('customer_state')['late_pct'].agg(['mean', 'std']).reset_index()
state_stats.columns = ['customer_state', 'mean_late_pct', 'std_late_pct']

# Merge back onto monthly data
monthly_filtered = monthly_filtered.merge(state_stats, on='customer_state')

# Calculate Z-score for each month (how many standard deviations above normal)
monthly_filtered['z_score'] = (
    (monthly_filtered['late_pct'] - monthly_filtered['mean_late_pct']) /
    monthly_filtered['std_late_pct']
)

#Flag anomalies (z-score > 2)
monthly_filtered['is_anomaly'] = monthly_filtered['z_score'] > 2

anomalies = monthly_filtered[monthly_filtered['is_anomaly']].sort_values('z_score', ascending=False)
print(f"\nAnomalies detected: {len(anomalies)}")
print(anomalies[['customer_state', 'order_month', 'late_pct', 'mean_late_pct', 'z_score']].to_string())
```

```
States included: ['BA', 'CE', 'DF', 'ES', 'GO', 'MA', 'MG', 'MS', 'MT', 'PA', 'PE', 'PR', 'RJ', 'RS', 'SC', 'SP']

Anomalies detected: 22
    customer_state order_month  late_pct  mean_late_pct   z_score
315             SP    2016-09    100.00       9.440909  4.411785
141             MG    2018-03     23.25       4.143333  3.470944
78              ES    2018-03     44.36       9.750476  3.437955
203             PA    2018-03     52.24       9.761429  3.362588
170             MT    2017-03     37.50       6.595238  3.202502
246             PR    2018-03     15.18       3.936818  2.739407
288             RS    2018-03     20.64       5.632381  2.691776
36              CE    2018-03     53.76      12.836190  2.665344
161             MS    2018-03     40.35       9.000500  2.630859
15              BA    2018-03     35.32      11.929048  2.629585
98              GO    2018-02     21.92       6.377143  2.529905
120             MA    2018-03     55.10      16.805238  2.410294
57              DF    2018-03     20.41       5.762857  2.398856
116             MA    2017-11     54.72      16.805238  2.386376
166             MS    2018-08     37.14       9.000500  2.361475
267             RJ    2018-03     36.92      10.426190  2.358759
308             SC    2018-02     25.98       8.471429  2.351499
266             RJ    2018-02     35.95      10.426190  2.272399
224             PE    2018-03     23.36       9.500000  2.252625
99              GO    2018-03     19.86       6.377143  2.194600
182             MT    2018-03     26.92       6.595238  2.106151
309             SC    2018-03     23.48       8.471429  2.015735
```

```
# Plotting the graph

# CHART 5: Anomaly Detection Heatmap

# Pivot data for heatmap
# Convert order_month to string for plotting
monthly_filtered['order_month_str'] = monthly_filtered['order_month'].astype(str)

heatmap_data = monthly_filtered.pivot_table(
    index='customer_state',
    columns='order_month_str',
    values='z_score',
    aggfunc='mean'
)

# Sort states by their max z-score (worst anomalies at top)
heatmap_data = heatmap_data.loc[heatmap_data.max(axis=1).sort_values(ascending=False).index]

fig, ax = plt.subplots(figsize=(14, 7))

sns.heatmap(
    heatmap_data,
    cmap='RdYlGn_r',  # Red = bad, Green = good
    center=0,
    vmin=-2,
    vmax=4,
    ax=ax,
    linewidths=0.3,
    linecolor='gray',
    cbar_kws={'label': 'Z-Score (Standard Deviations Above Normal)'}
)

ax.set_title('Delivery Anomaly Heatmap by State and Month\n(Red = Abnormally High Late Rate | March 2018 = Systemic Failure)',
             fontsize=13, fontweight='bold')
ax.set_xlabel('Month', fontsize=11)
ax.set_ylabel('State', fontsize=11)
plt.xticks(rotation=45, ha='right', fontsize=8)
plt.yticks(fontsize=9)

plt.tight_layout()
plt.savefig('chart5_anomaly_heatmap.png', bbox_inches='tight')
plt.show()
```




```
# CHART 6: March 2018 Spike — Before vs During

# Focus on top 8 states by volume
top_states = ['SP', 'RJ', 'MG', 'RS', 'PR', 'SC', 'BA', 'CE']

march_comparison = monthly_filtered[
    monthly_filtered['customer_state'].isin(top_states)
][['customer_state', 'order_month_str', 'late_pct', 'mean_late_pct']].copy()

# Filter to just Feb, Mar, Apr 2018 to show the spike
spike_window = march_comparison[
    march_comparison['order_month_str'].isin(['2018-01', '2018-02', '2018-03', '2018-04', '2018-05'])
]

fig, ax = plt.subplots(figsize=(10, 5))

for state in top_states:
    state_data = spike_window[spike_window['customer_state'] == state]
    ax.plot(state_data['order_month_str'], state_data['late_pct'],
            marker='o', label=state, linewidth=2)

ax.axvline(x='2018-03', color='red', linestyle='--', linewidth=2, label='March 2018 (Anomaly)')
ax.set_title('Late Delivery Rate Jan–May 2018\nThe March 2018 Systemic Failure',
             fontsize=13, fontweight='bold')
ax.set_xlabel('Month')
ax.set_ylabel('Late Delivery Rate (%)')
ax.legend(bbox_to_anchor=(1.05, 1), loc='upper left')

plt.tight_layout()
plt.savefig('chart6_march2018_spike.png', bbox_inches='tight')
plt.show()
```



The Bonuc Challenge

```
# Load the translation file
translations_df = pd.read_csv('/content/drive/MyDrive/Amali/product_category_name_translation.csv')
print(translations_df.shape)
print(translations_df.head(10))
```

```
(71, 2)
   product_category_name product_category_name_english
0          beleza_saude                health_beauty
1  informatica_acessorios        computers_accessories
2            automotivo                      auto
3        cama_mesa_banho               bed_bath_table
4       moveis_decoracao             furniture_decor
5          esporte_lazer              sports_leisure
6            perfumaria                 perfumery
7   utilidades_domesticas                 housewares
8            telefonia                 telephony
9     relogios_presentes               watches_gifts
```

```
# Join translations to master_df
master_df = master_df.merge(translations_df, on='product_category_name', how='left')

# Check how many got translated vs still null
print(f"Total rows: {len(master_df)}")
print(f"Translated: {master_df['product_category_name_english'].notna().sum()}")
print(f"Still null: {master_df['product_category_name_english'].isna().sum()}")

# Fill remaining nulls (products with no category get labeled unknown)
master_df['product_category_name_english'] = master_df['product_category_name_english'].fillna('unknown')
```

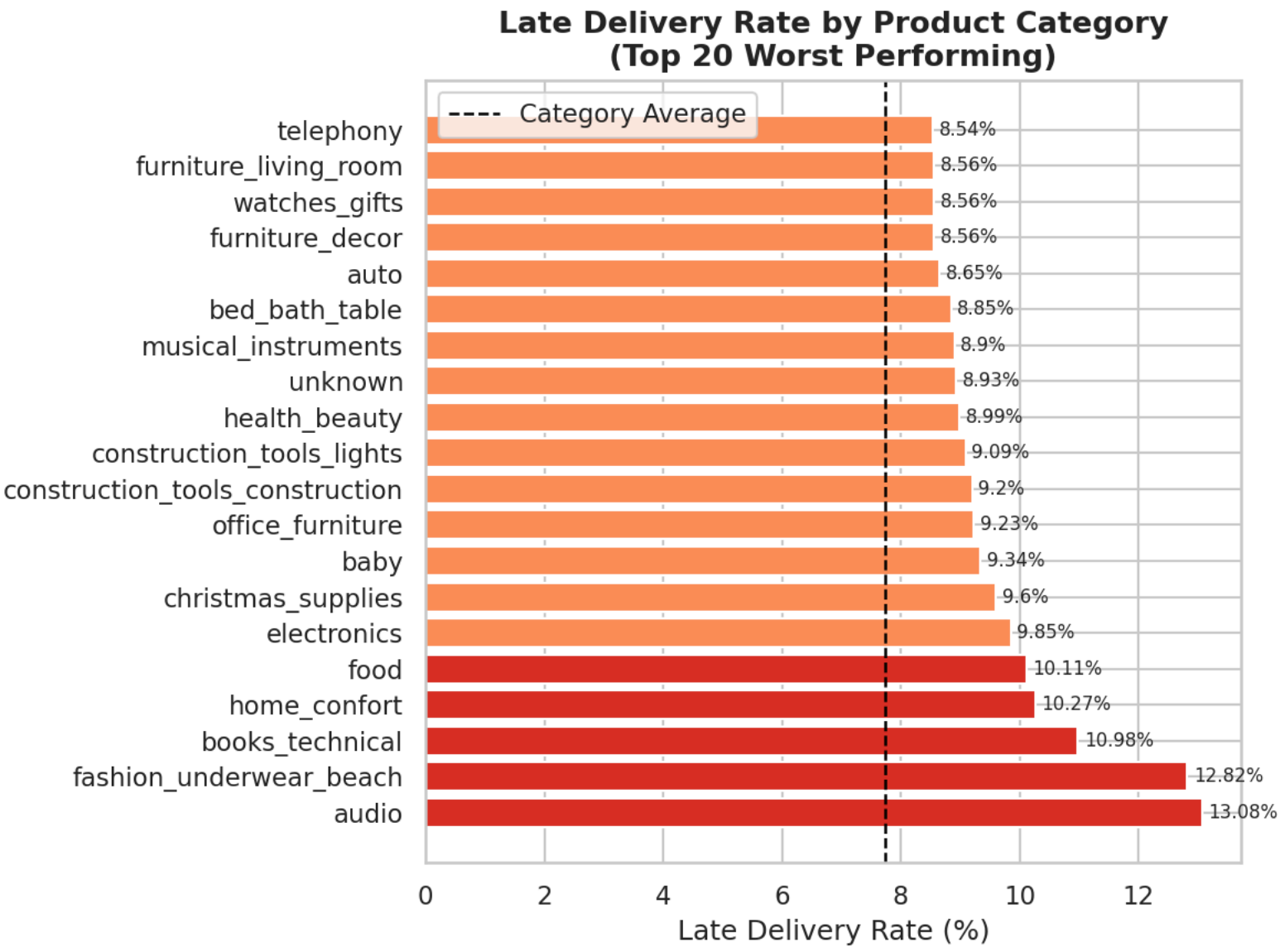
```
Total rows: 96470
Translated: 95092
Still null: 1378
```

```
# Analyze late rate per category
category_analysis = master_df.groupby('product_category_name_english').agg(
    total_orders=('order_id', 'count'),
    late_orders=('is_late', 'sum'),
    avg_review_score=('review_score', 'mean')
).reset_index()

category_analysis['late_pct'] = (
    category_analysis['late_orders'] / category_analysis['total_orders'] * 100
).round(2)

# Filter to categories with at least 100 orders
category_analysis = category_analysis[category_analysis['total_orders'] >= 100]

# Sort by worst late rate
category_analysis = category_analysis.sort_values('late_pct', ascending=False)

print(f"\nCategories with 100+ orders: {len(category_analysis)}")
print(category_analysis[['product_category_name_english', 'total_orders', 'late_pct', 'avg_review_score']].to_string())
```

```
Categories with 100+ orders: 52
           product_category_name_english  total_orders  late_pct  avg_review_score
4                                  audio           344     13.08          3.847507
33                fashion_underwear_beach           117     12.82          4.008621
10                      books_technical           255     10.98          4.425197
47                        home_confort            370     10.27          3.907609
36                                food            435     10.11          4.324826
26                         electronics          2507      9.85          4.128617
12                   christmas_supplies           125      9.60          4.113821
6                                 baby          2763      9.34          4.121489
57                     office_furniture          1246      9.23          3.650485
17         construction_tools_construction        728      9.20          4.132780
18              construction_tools_lights        231      9.09          4.214912
43                       health_beauty         8608      8.99          4.235342
70                             unknown         1378      8.93          4.023375
56                  musical_instruments          607      8.90          4.247920
7                        bed_bath_table         9167      8.85          4.010802
5                                  auto         3792      8.65          4.151033
39                      furniture_decor         6213      8.56          4.073131
71                        watches_gifts         5470      8.56          4.127072
40                 furniture_living_room          409      8.56          4.094059
68                            telephony         4076      8.54          4.056022
50          industry_commerce_and_business        228      8.33          4.234513
16                       consoles_games         1014      8.19          4.185075
48                     home_construction         466      8.15          4.053879
42                         garden_tools         3414      8.03          4.194109
2                                   art          191      7.85          4.126984
65                       sports_leisure         7490      7.80          4.233280
66                           stationery         2247      7.79          4.306177
15                 computers_accessories         6500      7.74          4.083939
69                                 toys         3778      7.52          4.240128
59                            perfumery         3073      7.42          4.266405
21              costruction_tools_garden         183      7.10          4.278689
45                   home_appliances_2          226      7.08          4.155556
49                          housewares         5688      7.00          4.203537
8                 books_general_interest         492      6.91          4.538934
20                          cool_stuff         3530      6.86          4.227584
14                           computers         176      6.82          4.242775
44                       home_appliances         747      6.69          4.216398
24                                drinks         284      6.69          4.257042
28               fashion_bags_accessories         1810      6.63          4.241953
31                        fashion_shoes          233      6.44          4.275109
37                           food_drink          218      6.42          4.444954
60                             pet_shop         1682      6.30          4.280167
63                     small_appliances          606      6.27          4.264559
62                 signaling_and_security          138      5.80          4.116788
53                    luggage_accessories         1008      5.46          4.379482
34                       fixed_telephony          212      5.19          3.966507
0              agro_industry_and_commerce          177      5.08          4.107345
54                          market_place          268      4.85          4.116541
30                  fashion_male_clothing          106      4.72          3.819048
51  kitchen_dining_laundry_garden_furniture          239      4.60          4.135021
19               construction_tools_safety          154      4.55          4.019608
1                       air_conditioning          245      4.08          4.128631
```

```
# CHART 7: Late Rate by Product Category

top20_categories = category_analysis.head(20)

fig, ax = plt.subplots(figsize=(8, 6))

colors = ['#d73027' if x > 10 else '#fc8d59' if x > 8 else '#91bfdb'
          for x in top20_categories['late_pct']]

bars = ax.barh(top20_categories['product_category_name_english'],
```

```
            top20_categories['late_pct'],
            color=colors)

ax.axvline(x=category_analysis['late_pct'].mean(),
         color='black', linestyle='--', linewidth=1.2, label='Category Average')

ax.set_xlabel('Late Delivery Rate (%)')
ax.set_title('Late Delivery Rate by Product Category\n(Top 20 Worst Performing)',
           fontsize=13, fontweight='bold')
ax.legend()

for bar, val in zip(bars, top20_categories['late_pct']):
    ax.text(val + 0.1, bar.get_y() + bar.get_height()/2,
          f'{val}%', va='center', fontsize=8)

plt.tight_layout()
plt.savefig('chart7_late_by_category.png', bbox_inches='tight')
plt.show()
```




```
# Removing duplicates again after translations join
print(f"Before deduplication: {master_df.shape}")

master_df = master_df.drop_duplicates(subset='order_id', keep='first')

print(f"After deduplication: {master_df.shape}")
```

```
Before deduplication: (96470, 26)
```